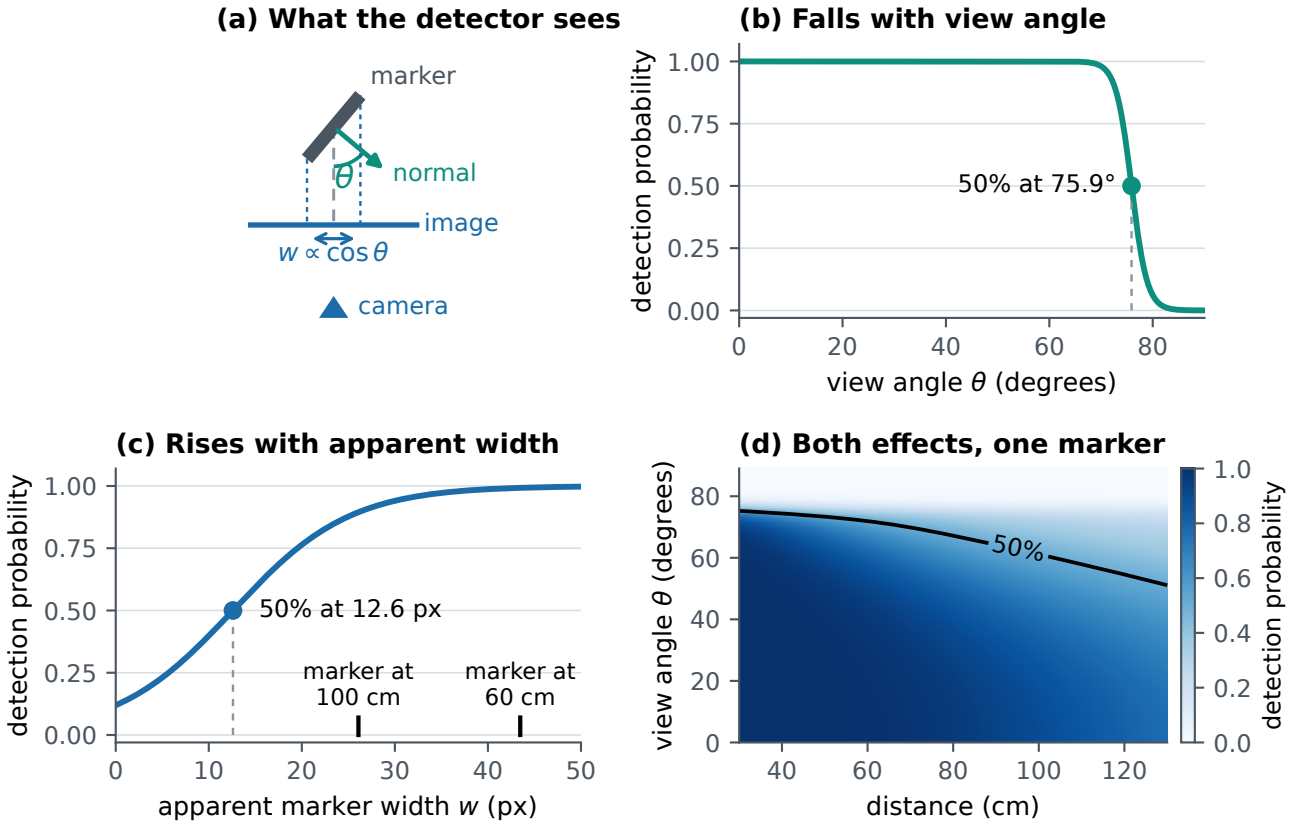

(a) What the detector sees
marker
normal
θ
image
w ∝ cos θ
camera
(b) Falls with view angle
detection probability
1.00
0.75
0.50
0.25
0.00
50% at 75.9°
0
20
40
60
80
view angle θ (degrees)
(c) Rises with apparent width
detection probability
1.00
0.75
0.50
0.25
0.00
50% at 12.6 px
marker at 100 cm
marker at 60 cm
0
10
20
30
40
50
apparent marker width w (px)
(d) Both effects, one marker
view angle θ (degrees)
80
60
40
20
0
50%
40
60
80
100
120
distance (cm)
1.0
0.8
0.6
0.4
0.2
0.0
detection probability